MAE saliency

Launcher: `Toolbox/mae.bat`

Purpose

This tool uses features from an ImageNet-pretrained Masked Autoencoder Vision Transformer (`vit_base_patch16_224.mae`) to create a saliency map. Higher values mean stronger model feature activation relative to other pixels in the same tile. They are not calibrated probabilities of archaeology or crop marks.

The implementation does not train an MAE and does not calculate reconstruction error. It measures the L2 magnitude of patch tokens from pretrained MAE features.

Inputs and parameters

Option	What to provide	Behaviour
Input folder	Folder containing `.tif` files	Only top-level files with the lowercase `.tif` extension are scanned; subfolders, `.tiff`, and uppercase extensions are not included. All eligible images in the folder are processed.
Bands to process	One to three comma-separated, one-based band numbers	Default: `1`. One number is repeated across all three model channels. Two numbers use the first band again as the third channel. Three numbers are used in the entered order. More than three numbers, non-integers, zero, and negative numbers are rejected.

Option	What to provide	Behaviour
Local MAE checkpoint	Optional `.safetensors`, `.bin`, `.pth`, or `.pt` file	Blank uses the cached `vit_base_patch16_224.mae` weights or downloads them on first use. Select a compatible local checkpoint for offline execution.
Vector layer	Optional clip polygon	Clips each completed saliency raster in place. Vector geometries are transformed to the raster CRS.

There is no output-folder control. Results always go to `mae_saliency` inside the input folder.

Band and tile behaviour

- Images are divided into 512×512-pixel tiles with a default overlap of 128 pixels. The final tile along each axis is shifted to the raster edge to retain full coverage and a consistent tile size whenever the raster is at least 512 pixels wide or high.
- The selected source bands form the model's three input channels. For example, `5,3,2` uses source bands 5, 3, and 2 in that order; 4 becomes `4,4,4`; and `4,2` becomes `4,2,4`.
- Every selected number must exist in every input image. Processing stops with an error naming the image and unavailable band if it does not.
- Each tile and each of its three channels is independently min-max normalized, then resized to 224×224 for the model.
- The model produces a 14×14 token-magnitude grid, which is bilinearly enlarged to the tile's original dimensions.
- Overlapping predictions are combined pixel by pixel as a weighted average using a raised two-dimensional Hann window. Its default minimum weight is `0.001`, which keeps normalization stable where only one tile contributes.
- Tile-wise normalization means absolute saliency values can still vary between tiles, although raised-Hann blending suppresses abrupt boundary artefacts. Values should be compared cautiously across images.
- The model uses CUDA when available and CPU otherwise. The pretrained weights are stored in the user's model cache rather than inside `env_mae`. They may be downloaded on first use when no local checkpoint is selected.

Outputs

For each source image:

- `mae_saliency/<source-name>_mae.tif`: single-band Float32 saliency map. Its metadata records the effective three-channel source-band order, model source, tile overlap, blending window, and minimum Hann weight.
- `mae_saliency/mae_stats.csv`: one row per completed map with mean, median, minimum, 90th and 95th percentiles, maximum, and standard deviation

Existing `_mae.tif` outputs are skipped. Source filenames already ending `_mae` are also skipped. Temporary tile directories are deleted after successful processing; an interrupted or failed run can leave a `<source-name>_temp_tiles` folder.

If you change the band selection or checkpoint, move or delete existing outputs before running again; otherwise those outputs remain skipped.

Interpretation

Use the 90th or 95th percentile in `mae_stats.csv` as a within-image display threshold, not as a universal archaeological cutoff. Natural RGB photographs were used for model pretraining, so false responses to texture, boundaries, clouds, shadows, and uncommon spectral composites are expected. Review high-saliency areas against the input, other products, and archaeological context.